

- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. 2022. [Progress measures for grokking via mechanistic interpretability](#). In *The Eleventh International Conference on Learning Representations*.
- Lorenzo Noci, Sotiris Anagnostidis, Luca Biggio, Antonio Orvieto, Sidak Pal Singh, and Aurelien Lucchi. 2022. [Signal Propagation in Transformers: Theoretical Perspectives and the Role of Rank Collapse](#). In *Advances in Neural Information Processing Systems*.
- OpenAI. 2024. New reasoning models: Openai o1-preview and o1-mini. <https://openai.com/research/o1-preview-and-o1-mini>.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. [Training language models to follow instructions with human feedback](#). *arxiv:2203.02155[cs]*.
- Keiran Paster, Marco Dos Santos, Zhangir Azerbayev, and Jimmy Ba. 2023. [OpenWebMath: An Open Dataset of High-Quality Mathematical Web Text](#). In *The Twelfth International Conference on Learning Representations*.
- William Peebles and Saining Xie. 2023. [Scalable Diffusion Models with Transformers](#). In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4195–4205.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language Models are Unsupervised Multitask Learners. *OpenAI*, page 24.
- Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. 2019. [Compressive Transformers for Long-Range Sequence Modelling](#). *arxiv:1911.05507[cs]*.
- Samyam Rajbhandari, Jeff Rasley, Olatunji Ruwase, and Yuxiong He. 2020. [ZeRO: Memory optimizations Toward Training Trillion Parameter Models](#). In *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 1–16.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. [High-Resolution Image Synthesis With Latent Diffusion Models](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2021. [WinoGrande: An adversarial winograd schema challenge at scale](#). *Commun. ACM*, 64(9):99–106.
- Victor Sanh, Albert Webson, Colin Raffel, Stephen Bach, Lintang Sutawika, Zaid Alyafeai, Antoine Chaffin, Arnaud Stiegler, Arun Raja, Manan Dey, M. Saiful Bari, Canwen Xu, Urmish Thakker, Shanya Sharma Sharma, Eliza Szczechla, Taewoon Kim, Gunjan Chhablani, Nihal Nayak, Debajyoti Datta, and 21 others. 2021. [Multitask Prompted Training Enables Zero-Shot Task Generalization](#). In *International Conference on Learning Representations*.
- Sunny Sanyal, Atula Tejaswi Neerkaje, Jean Kaddour, Abhishek Kumar, and sujay sanghavi. 2024. [Early weight averaging meets high learning rates for LLM pre-training](#). In *First Conference on Language Modeling*.
- Juergen Schmidhuber. 2012. [Self-Delimiting Neural Networks](#). *arxiv:1210.0118[cs]*.
- Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. 2022. [Confident Adaptive Language Modeling](#). In *Advances in Neural Information Processing Systems*.
- Avi Schwarzschild. 2023. [Deep Thinking Systems: Logical Extrapolation with Recurrent Neural Networks](#). Ph.D. thesis, University of Maryland, College Park, College Park.
- Avi Schwarzschild, Eitan Borgnia, Arjun Gupta, Arpit Bansal, Zeyad Emam, Furong Huang, Micah Goldblum, and Tom Goldstein. 2021a. [Datasets for Studying Generalization from Easy to Hard Examples](#). *arxiv:2108.06011[cs]*.
- Avi Schwarzschild, Eitan Borgnia, Arjun Gupta, Furong Huang, Uzi Vishkin, Micah Goldblum, and Tom Goldstein. 2021b. [Can You Learn an Algorithm? Generalizing from Easy to Hard Problems with Recurrent Networks](#). In *Advances in Neural Information Processing Systems*, volume 34, pages 6695–6706. Curran Associates, Inc.
- Avi Schwarzschild, Sean Michael McLeish, Arpit Bansal, Gabriel Diaz, Alex Stein, Aakash Chandnani, Aniruddha Saha, Richard Baraniuk, Long Tran-Thanh, Jonas Geiping, and Tom Goldstein. 2023. [Algorithm Design for Learned Algorithms](#).
- Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. [Neural Machine Translation of Rare Words with Subword Units](#). In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1715–1725, Berlin, Germany. Association for Computational Linguistics.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Y Wu, and 1 others. 2024. [Deepseekmath: Pushing the limits of mathematical reasoning in open language models](#). *arXiv preprint arXiv:2402.03300*.
- Noam Shazeer. 2020. [GLU Variants Improve Transformer](#). *arxiv:2002.05202[cs]*.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. 2017. [Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer](#). *arxiv:1701.06538[cs]*.
- Siddharth Singh and Abhinav Bhatle. 2022. [AxiNN: An asynchronous, message-driven parallel framework for extreme-scale deep learning](#). In *2022 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 606–616.
- Siddharth Singh, Prajwal Singhania, Aditya Ranjan, John Kirchenbauer, Jonas Geiping, Yuxin Wen, Neel Jain, Abhimanyu Hans, Manli Shu, Aditya Tomar, Tom Goldstein, and Abhinav Bhatle. 2024. [Democratizing AI: Open-source Scalable LLM Training on GPU-based Supercomputers](#). In *2024 SC24: International Conference for High Performance Computing, Networking, Storage and Analysis SC*, pages 36–49. IEEE Computer Society.
- Oscar Skea, Md Rifat Arefin, Yann LeCun, and Ravid Shwartz-Ziv. 2024. [Does Representation Matter? Exploring Intermediate Layers in Large Language Models](#). *arxiv:2412.09563[cs]*.

- Daria Soboleva, Faisal Al-Khateeb, Joel Hestness, Nolan Dey, Robert Myers, and Jacob Robert Steeves. 2023. [SlimPajama: A 627B token cleaned and deduplicated version of RedPajama](#).
- Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, Jennifer Dumas, Yanai Elazar, Valentin Hofmann, Ananya Jha, Sachin Kumar, Li Lucy, Xinxu Lyu, Nathan Lambert, Ian Magnusson, Jacob Morrison, Niklas Muennighoff, and 17 others. 2024. [Dolma: An Open Corpus of Three Trillion Tokens for Language Model Pretraining Research](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15725–15788, Bangkok, Thailand. Association for Computational Linguistics.
- Yang Song and Stefano Ermon. 2019. [Generative Modeling by Estimating Gradients of the Data Distribution](#). *arXiv:1907.05600 [cs, stat]*.
- Jianlin Su, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. 2021. [RoFormer: Enhanced Transformer with Rotary Position Embedding](#). *arxiv:2104.09864 [cs]*.
- Sainbayar Sukhbaatar, Edouard Grave, Guillaume Lample, Herve Jegou, and Armand Joulin. 2019. [Augmenting Self-attention with Persistent Memory](#). *arxiv:1907.01470 [cs, stat]*.
- Qi Sun, Marc Pickett, Aakash Kumar Nain, and Llion Jones. 2024. [Transformer Layers as Painters](#). *arxiv:2407.09298 [cs]*.
- Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei Efros, and Moritz Hardt. 2020. [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#). In *Proceedings of the 37th International Conference on Machine Learning*, pages 9229–9248. PMLR.
- Ilya Sutskever, Geoffrey E Hinton, and Graham W Taylor. 2008. [The Recurrent Temporal Restricted Boltzmann Machine](#). In *Advances in Neural Information Processing Systems*, volume 21. Curran Associates, Inc.
- Mirac Suzgun, Sebastian Gehrmann, Yonatan Belinkov, and Stuart M. Shieber. 2019. [Memory-Augmented Recurrent Neural Networks Can Learn Generalized Dyck Languages](#). *arxiv:1911.03329 [cs]*.
- Sho Takase and Shun Kiyono. 2023. [Lessons on Parameter Sharing across Layers in Transformers](#). *arxiv:2104.06022 [cs]*.
- Sho Takase, Shun Kiyono, Sosuke Kobayashi, and Jun Suzuki. 2024. [Spike No More: Stabilizing the Pre-training of Large Language Models](#). *arxiv:2312.16903 [cs]*.
- Shawn Tan, Yikang Shen, Zhenfang Chen, Aaron Courville, and Chuang Gan. 2023. [Sparse Universal Transformer](#). *arxiv:2310.07096 [cs]*.
- Team Gemma, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabella Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, and 179 others. 2024. [Gemma 2: Improving Open Language Models at a Practical Size](#). *arxiv:2408.00118 [cs]*.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, and 21 others. 2025. [2 OLMo 2 Furious](#). *arxiv:2501.00656 [cs]*.
- TogetherAI. 2023. [Llama-2-7B-32K-Instruct — and fine-tuning for Llama-2 models with Together API](#).
- Shubham Toshniwal, Wei Du, Ivan Moshkov, Branislav Kısacanin, Alexan Ayrapetyan, and Igor Gitman. 2024a. [OpenMathInstruct-2: Accelerating AI for Math with Massive Open-Source Instruction Data](#). *arxiv:2410.01560 [cs]*.
- Shubham Toshniwal, Ivan Moshkov, Sean Narenthiran, Daria Gitman, Fei Jia, and Igor Gitman. 2024b. [OpenMathInstruct-1: A 1.8 Million Math Instruction Tuning Dataset](#). In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. [Attention Is All You Need](#). *arXiv:1706.03762 [cs]*.
- Zengzhi Wang, Xuefeng Li, Rui Xia, and Pengfei Liu. 2024a. [MathPile: A Billion-Token-Scale Pretraining Corpus for Math](#). In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Zhilin Wang, Yi Dong, Olivier Delalleau, Jiaqi Zeng, Gerald Shen, Daniel Egert, Jimmy J. Zhang, Makesh Narsimhan Sreedhar, and Oleksii Kuchaiev. 2024b. [HelpSteer2: Open-source dataset for training top-performing reward models](#). *arxiv:2406.08673 [cs]*.
- Maurice Weber, Daniel Y. Fu, Quentin Gregory Anthony, Yonatan Oren, Shane Adams, Anton Alexandrov, Xiaozhong Lyu, Huu Nguyen, Xiaozhe Yao, Virginia Adams, Ben Athiwaratkun, Rahul Chalamala, Kezhen Chen, Max Ryabinin, Tri Dao, Percy Liang, Christopher Re, Irina Rish, and Ce Zhang. 2024. [RedPajama: An Open Dataset for Training Large Language Models](#). In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Ronald J. Williams and Jing Peng. 1990. [An Efficient Gradient-Based Algorithm for On-Line Training of Recurrent Network Trajectories](#). *Neural Computation*, 2(4):490–501.
- Mitchell Wortsman, Tim Dettmers, Luke Zettlemoyer, Ari Morcos, Ali Farhadi, and Ludwig Schmidt. 2023a. [Stable and low-precision training for large-scale vision-language models](#). *Advances in Neural Information Processing Systems*, 36:10271–10298.
- Mitchell Wortsman, Tim Dettmers, Luke Zettlemoyer, Ari S. Morcos, Ali Farhadi, and Ludwig Schmidt. 2023b. [Stable and low-precision training for large-scale vision-language models](#). In *Thirty-Seventh Conference on Neural Information Processing Systems*.
- Mengshiou Wu and Mark Stock. 2024. [Enhancing PyTorch Performance on Frontier with the RCCL OFI-Plugin](#).
- Yuhuai Wu, Markus Norman Rabe, DeLesley Hutchins, and Christian Szegedy. 2022. [Memorizing Transformers](#). In *International Conference on Learning Representations*.

- Zijian Wu, Jiayu Wang, Dahua Lin, and Kai Chen. 2024. [LEAN-GitHub: Compiling GitHub LEAN repositories for a versatile LEAN prover](#). *arxiv:2407.17227[cs]*.
- Zhangchen Xu, Fengqing Jiang, Luyao Niu, Yuntian Deng, Radha Poovendran, Yejin Choi, and Bill Yuchen Lin. 2024. [Magpie: Alignment Data Synthesis from Scratch by Prompting Aligned LLMs with Nothing](#). *arxiv:2406.08464[cs]*.
- Kaiyu Yang, Aidan M. Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, and Anima Anandkumar. 2023. [LeanDojo: Theorem Proving with Retrieval-Augmented Language Models](#). In *Thirty-Seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Liu Yang, Kangwook Lee, Robert D. Nowak, and Dimitris Papailiopoulos. 2024a. [Looped Transformers are Better at Learning Learning Algorithms](#). In *The Twelfth International Conference on Learning Representations*.
- Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. 2024b. [Parallelizing Linear Transformers with the Delta Rule over Sequence Length](#). In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.
- Huayuan Ying, Zijian Wu, Yihan Geng, Jiayu Wang, Dahua Lin, and Kai Chen. 2024. [Lean Workbook: A large-scale Lean problem set formalized from natural language math problems](#). In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. 2023. [MetaMath: Bootstrap Your Own Mathematical Questions for Large Language Models](#). In *The Twelfth International Conference on Learning Representations*.
- Pedram Zamirai, Jian Zhang, Christopher R. Aberger, and Christopher De Sa. 2021. [Revisiting BFloat16 Training](#). *arxiv:2010.06192[cs, stat]*.
- Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D. Goodman. 2024. [Quiet-STaR: Language Models Can Teach Themselves to Think Before Speaking](#). *arxiv:2403.09629[cs]*.
- Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.
- Xiaohua Zhai, Alexander Kolesnikov, Neil Houlsby, and Lucas Beyer. 2022. [Scaling Vision Transformers](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12104–12113.
- Biao Zhang and Rico Sennrich. 2019. [Root Mean Square Layer Normalization](#). In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.
- Ge Zhang, Scott Qu, Jiaheng Liu, Chenchen Zhang, Chenghua Lin, Chou Leuang Yu, Danny Pan, Esther Cheng, Jie Liu, Qunshu Lin, Raven Yuan, Tuney Zheng, Wei Pang, Xinrun Du, Yiming Liang, Yinghao Ma, Yizhi Li, Ziyang Ma, Bill Lin, and 26 others. 2024a. [MAP-Neo: Highly Capable and Transparent Bilingual Large Language Model Series](#). *arxiv:2405.19327[cs]*.
- Jun Zhang, Jue Wang, Huan Li, Lidan Shou, Ke Chen, Gang Chen, and Sharad Mehrotra. 2024b. [Draft& Verify: Lossless Large Language Model Acceleration via Self-Speculative Decoding](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11263–11282, Bangkok, Thailand. Association for Computational Linguistics.
- Yifan Zhang, Yifan Luo, Yang Yuan, and Andrew C. Yao. 2024c. [Autonomous Data Selection with Language Models for Mathematical Texts](#). In *ICLR 2024 Workshop on Navigating and Addressing Data Problems for Foundation Models*.
- Tianyu Zheng, Ge Zhang, Tianhao Shen, Xueling Liu, Bill Yuchen Lin, Jie Fu, Wenhui Chen, and Xiang Yue. 2024. [OpenCodeInterpreter: Integrating Code Generation with Execution and Refinement](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 12834–12859, Bangkok, Thailand. Association for Computational Linguistics.
- Fan Zhou, Zengzhi Wang, Qian Liu, Junlong Li, and Pengfei Liu. 2024. [Programming Every Example: Lifting Pre-training Data Quality like Experts at Scale](#). *arxiv:2409.17115[cs]*.
- Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widyasari, Imam Nur Bani Yusuf, Haolan Zhan, Junda He, Indraneil Paul, Simon Brunner, Chen Gong, Thong Hoang, Armel Randy Zebaze, Xiaoheng Hong, Wen-Ding Li, Jean Kaddour, Ming Xu, Zhihan Zhang, and 14 others. 2024. [BigCodeBench: Benchmarking Code Generation with Diverse Function Calls and Complex Instructions](#).